

Vignesh Sivakumar

vigneshsiva076@gmail.com · +91 81244 33423 · LinkedIn · GitHub · Portfolio

Summary

Full Stack Engineer with hands-on experience building production-grade, scalable web systems using **React**, **Node.js**, and **Java**. Open-source contributor to **Apache Arrow-RS** (merged PR #9167). Proven ability to optimise performance-critical UIs (60fps grid rendering, 50+ concurrent WebSocket connections) and deliver AI-integrated features. Passionate about clean architecture, RESTful API design, and algorithmic problem-solving. Seeking a Software Engineer role to build impactful, enterprise-scale systems.

Technical Skills

Languages:	C, Java, JavaScript (ES2022+)
Frontend:	React.js, Typescript, HTML5, CSS3, Tailwind CSS, Sass
Backend:	Node.js, Express.js, REST APIs, WebSockets (Socket.io), Middleware Design
Databases:	MySQL, MongoDB
AI/ML APIs:	Google Gemini, OpenAI API, Hugging Face Inference API
Tools:	Git, GitHub Actions (CI/CD), Docker (basic), Postman, Figma, Jira
Concepts:	Data Structures & Algorithms, System Design (Scalability, Caching), Async Programming

Projects

AI-Powered Real-Time Chat Application

GitHub

React.js · Node.js · Socket.io · Google Gemini API · JWT · MongoDB

- Engineered a production-ready, bidirectional messaging platform sustaining **50+ concurrent connections** with Socket.io; implemented room-based architecture and event-driven message routing.
- Integrated **Google Gemini AI** features (summarisation, smart reply, real-time translation, voice-to-text) reducing manual user effort by **~60%** in multi-language conversations.
- Implemented **JWT-based authentication**, server-side content moderation, and intelligent caching strategies that improved API response latency and strengthened overall application security.

Classless — Inclusive AI Tutor

GitHub

React.js · Node.js · Express.js · Hugging Face · REST APIs

- Designed a multilingual React UI supporting 5+ languages, optimized for low-bandwidth environments with lazy loading and minimal bundle size.
- Integrated AI tutoring via **OpenAI** and **Hugging Face** APIs; engineered adaptive UI flows optimised for **low-bandwidth environments**, broadening accessibility for underserved learners.
- Developed **RESTful backend APIs** using Node.js and Express.js; validated all integration flows with Postman and maintained comprehensive component documentation for long-term maintainability.

Pathfinding Algorithm Visualiser

GitHub

React.js · Asynchronous JavaScript · CSS Animations

- Built an interactive, animation-heavy web application using React.js showcasing **6+ pathfinding algorithms** (BFS, DFS, Dijkstra, A*, etc.) with cross-browser compatible, optimised grid rendering.
- Implemented **play/pause/rewind controls** and timeline scrubbing; added performance analytics (visited nodes count, execution time, shortest-path length) to quantify algorithm efficiency.

Open Source Contributions

Apache Arrow-RS

PR #9167 — Merged

Rust · Parquet · Documentation Engineering

- Improved developer experience for the **Apache Arrow-RS** Parquet crate by updating async **ArrowReaderBuilder** documentation with accurate, production-tested usage examples.
- Navigated a structured **open-source PR review workflow** with project maintainers; changes successfully merged into the main branch, contributing to a widely used data engineering library.

Education

Bannari Amman Institute of Technology

Sept 2023 – July 2027

B.E. in Information Science and Engineering — CGPA: 7.41 / 10.0

Tamil Nadu, India